

10(a)(i)	<u>molecules / moles / volume, carbon dioxide produced</u> molecules / moles / volume, oxygen taken in ;	1
10(a)(ii)	25 ;	1
10(a)(iii)	0.72 ; A ecf 18 divided by their answer to 10(a)(ii)	1

Question	Answer	Marks														
10(b)	any four from: 1 RQ (initially) at 1.0 as, carbohydrates / (named) sugar, respired / metabolised ; 2 RQ decreases (over time) as, carbohydrates / (named) sugar, running out / used up ; 3 RQ decreases (over time) as a mixture of carbohydrate and lipid respired ; I proteins 4 RQ at, 0.65 / 0.7, as, fatty acids / fats / lipids, respired (as main respiratory substrate) or from 120 min onwards, fatty acids / fats / lipids, respired (as main respiratory substrate) ; 5 data quote ; <i>RQ of 1.0 at 40 min plus any RQ value at any other time</i> <table><tr><th>Time after feeding / min</th><th>RQ ratio</th></tr><tr><td>40</td><td>1.0</td></tr><tr><td>80</td><td>0.75</td></tr><tr><td>120</td><td>0.7</td></tr><tr><td>160</td><td>0.65</td></tr><tr><td>200</td><td>0.7</td></tr><tr><td>240</td><td>0.7</td></tr></table>	Time after feeding / min	RQ ratio	40	1.0	80	0.75	120	0.7	160	0.65	200	0.7	240	0.7	4
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